

Note

Configurations Arising from Maximal Arcs

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We show that the existence of a maximal arc in a finite projective plane implies the existence of certain block designs and partial geometries. The block designs obtained have interesting resolvability properties.

We shall show that the existence of various combinatorial configurations is a consequence of the existence of a maximal arc in a finite projective plane. We shall start by defining our terms.

Suppose $\mathcal{P}$ is a finite projective plane of order s (that is, there are $s + 1$ points on each line). By a *maximal arc* $\mathcal{A}$ of $\mathcal{P}$ of parameter n ($0 < n < s + 1$) or *maximal n -arc*, we shall mean a set $\mathcal{A}$ of a points of $\mathcal{P}$ such that each line of $\mathcal{P}$ is incident either with n points or with no points of $\mathcal{A}$.

A *balanced incomplete block design* is a method of selecting b k -sets (blocks) from a set v objects (treatments) such that each treatment belongs to r blocks and each pair of distinct treatments occur together in λ blocks. A balanced incomplete block design is called *resolvable* if its blocks can be partitioned into r disjoint sets of br^{-1} blocks in such a way that every treatment belongs to precisely one block in each set. Such a set, in which all the blocks are disjoint, is called a *parallel class*.

A *partial geometry* (r, k, t) is a set $\mathcal{G}$ of points and a set $\Lambda(\mathcal{G})$ of lines such that

- (i) any two points lie on at most one line,
- (ii) any point lies on r lines,
- (iii) any line-passes through k points,

if $P \in \mathcal{G}$ does not lie on $l \in \Lambda(\mathcal{G})$, then there are exactly t lines through P intersecting l .

Maximal arcs are discussed in [3], while the properties of partial geometries may be found in [2].

THEOREM. *If there is a maximal arc of parameter n in a projective plane of order s , then there exist balanced incomplete block designs of parameters*

$$(i) \quad v = ns - s + n, \quad b = n^{-1}(ns - s + n)(s + 1), \\ r = s + 1, \quad k = n, \quad \lambda = 1$$

and

$$(ii) \quad v = n^{-1}s(s - n + 1), \quad b = (s - n + 1)(s + 1), \\ r = s + 1, \quad k = n^{-1}s, \quad \lambda = 1.$$

The designs are resolvable, and in fact any two disjoint blocks can be extended to a parallel class of br^{-1} blocks.

Proof. Suppose $\mathcal{A}$ is a maximal arc in $\mathcal{P}$, as defined above. Then clearly

$$a = (n - 1)(s - 1) + 1 = ns - s + n. \quad (1)$$

Let us write $\Lambda'(\mathcal{A})$ for the set of all lines of $\mathcal{P}$ which are incident with the members of $\mathcal{A}$. To determine the size of $\Lambda'(\mathcal{A})$ we shall count the number π of pairs (l, P) where $P \in \mathcal{A}$ and l is a line through P . Since there are a points in $\mathcal{A}$ and $s + 1$ lines through each point,

$$\pi = a(s + 1);$$

on the other hand each of the $|\Lambda'(\mathcal{A})|$ lines contains n points of $\mathcal{A}$, so

$$\pi = n |\Lambda'(\mathcal{A})|;$$

comparing

$$|\Lambda'(\mathcal{A})| = an^{-1}(s + 1). \quad (2)$$

Write $\mathcal{B}$ and $\Lambda(\mathcal{B})$ for the set-theoretic complements in $\mathcal{P}$ of $\mathcal{A}$ and $\Lambda'(\mathcal{A})$ respectively. We will need to know how many members of $\Lambda(\mathcal{B})$ contain a given member Q of $\mathcal{B}$. There is a line joining Q to each point of $\mathcal{A}$; each of these lines contains n points of $\mathcal{A}$, so there are an^{-1} distinct lines. These lines belong to $\Lambda'(\mathcal{A})$, and every other line through Q is in $\Lambda(\mathcal{B})$. So the number required is

$$s + 1 - an^{-1} \quad (3)$$

An immediate consequence is that n must divide a ; hence, using (1), n must divide s . (This was already known—see [1].)

Let us denote by $\Lambda(\mathcal{A})$ the set whose members are the intersections of elements of $\Lambda'(\mathcal{A})$ with $\mathcal{A}$. This set has $|\Lambda'(\mathcal{A})|$ elements. Each element contains n points, and each point of $\mathcal{A}$ belongs to $s + 1$ members of $\Lambda(\mathcal{A})$.

If we take any two points of $\mathcal{A}$ then they determine one line of $\mathcal{P}$, and that line is in $\Lambda'(\mathcal{A})$. So the two points lie in the corresponding member of $\Lambda(\mathcal{A})$, and only in that member. (If they were in two members, then two lines of $\Lambda'(\mathcal{A})$ would have two points in common.) Consequently, if we consider the members of $\mathcal{A}$ as treatments and the members of $\Lambda(\mathcal{A})$ as blocks, we have a balanced incomplete block design with $b = |\Lambda'(\mathcal{A})|$, $v = |\mathcal{A}|$, $r = s + 1$, $k = n$, $\lambda = 1$. But $|\mathcal{A}| = a$, and using (1) and (2) we see that this block design has the parameters required in part (i) of the Theorem.

Similarly we may consider the sets $\mathcal{B}$ and $\Lambda(\mathcal{B})$. In this case any two lines in $\Lambda(\mathcal{B})$ determine a point of $\mathcal{B}$, but two points of $\mathcal{B}$ need not determine a line in $\Lambda(\mathcal{B})$; so we take $\Lambda(\mathcal{B})$ as the set of treatments, and with each point P of $\mathcal{B}$ associate the block whose elements are the lines passing through P . Then every treatment belongs to $s + 1$ blocks and the size of the blocks is given by (3). We again have a block design. Using the facts

$$\begin{aligned} |\mathcal{B}| &= s^2 + s + 1 - |\mathcal{A}| \\ |\Lambda(\mathcal{B})| &= s^2 + s + 1 - |\Lambda'(\mathcal{A})| \end{aligned}$$

and simplifying the resulting parameters with (1), we find that the design is the one required for (ii).

To observe that the designs are resolvable, we select a line l disjoint from $\mathcal{A}$. If m_1 and m_2 are any two lines of $\Lambda'(\mathcal{A})$ which intersect l in the same point P , then m_1 and m_2 have no common point except P , so $m_1 \cap \mathcal{A}$ and $m_2 \cap \mathcal{A}$ have no common point. In this way we see that the set of all lines through P , when intersected with $\mathcal{A}$, give a parallel class of blocks in the design (i); by considering all points P on l we see that the design is resolvable. In fact, suppose m_1 and m_2 are any two lines of $\Lambda'(\mathcal{A})$ whose point of intersection, P say, is not in $\mathcal{A}$. There are $s + 1$ lines through P and of these the number which contain points of $\mathcal{A}$ is $n^{-1}(ns - s + n)$, so that at least $n^{-1}s$ of the lines do not contain any point of $\mathcal{A}$. If we choose one of these lines as l , and proved as above, we obtain a resolution in which $m_1 \cap \mathcal{A}$ and $m_2 \cap \mathcal{A}$ lie in the same parallel class. Since $m_1 \cap \mathcal{A}$ and $m_2 \cap \mathcal{A}$ were general non-intersecting members of $\Lambda(\mathcal{A})$, we have shown that any two blocks of design (i) with empty intersection can be extended to a parallel class of blocks.

It is easy to prove that these latter results also hold for the design (ii). So we have the Theorem.

Remark. The balanced incomplete block designs of the Theorem were shown to exist in [5] and [6] for the special case where $n = 2$.

COROLLARY. *If there is a maximal arc of parameter n in a projective*

plane of order s , then there exist partial geometries with the following parameters:

- (i) $(n, s + 1, n)$
- (ii) $(s + 1, n, n)$
- (iii) $(n^{-1}s, s + 1, n^{-1}s)$
- (iv) $(s + 1, n^{-1}s, n^{-1}s)$
- (v) $(s + 1 - n^{-1}s, s - n + 1, s + 1 - n - n^{-1}s)$.

Proof. Bose [2] has pointed out that a block design with $\lambda = 1$ is a partial geometry (r, k, k) under the interpretation of blocks as lines and treatments as points. Moreover, the dual of such a partial geometry (reinterpreting points as lines and lines as points) is a partial geometry (k, r, k) . Hence (i)–(iv) follow immediately from the Theorem. However, it is interesting to observe that in (i) and (iv) the set $\mathcal{S}$ (of definition (c)) is the set $\mathcal{A}$ and is the set $\mathcal{B}$ respectively, so the partial geometry has arisen in a natural way as sets of points and lines in a projective plane. The dual of $\Delta(\mathcal{B})$ forms a maximal arc of parameter $n^{-1}s$ in the dual plane of $\mathcal{P}$, so (ii) and (iii) arise naturally also.

The partial geometry (v) can be obtained by taking $\mathcal{S}$ as the set of points and taking the intersections of members of $\Delta'(\mathcal{A})$ with $\mathcal{B}$ as the lines.

Suppose we have a block design with the parameters of part (ii) of the Theorem. We may ask whether this design arises from a maximal arc in some projective plane. It will not always do so. For example, the case $n = 3, s = 9$ yields parameters

$$v = 21, \quad b = 70, \quad r = 10, \quad k = 3, \quad \lambda = 1.$$

If we take the design to have blocks

$$\begin{aligned} &\{i, 1 + i, 13 + i\}(\bmod 21) \\ &\{i, 4 + i, 10 + i\}(\bmod 21) \quad (0 \leq i \leq 20) \\ &\{i, 16 + i, 19 + i\}(\bmod 21) \\ &\{i, 7 + i, 14 + i\} \quad (0 \leq i \leq 6) \end{aligned}$$

(taken from p. 293 of [4]), then it does not arise from any maximal arc; in fact, a calculation shows that the disjoint blocks $\{0, 1, 13\}$ and $\{3, 4, 16\}$ cannot be extended to a parallel class of blocks.

REFERENCES

1. A. BARLOTTI, Sui $(k; n)$ — Archi di un Piano Lineare Finito, *Boll. Un. Mat. Ital.* **11** (1956), 553–556.
2. R. C. BOSE, Strongly Regular Graphs, Partial Geometries and Partially Balanced Designs, *Pacific J. Math.* **13** (1963), 389–419.
3. R. H. F. DENNISTON, Some Maximal Arcs in Finite Projective Planes, *J. Combinatorial Theory* **6** (1969), 317–319.
4. M. HALL, JR., "Combinatorial Theory," Blaisdell, Waltham, Mass., 1967.
5. E. SEIDEN, On a method of construction of partial geometries and partial Bolyai-Lobachevsky planes, *Amer. Math. Monthly* **73** (1966), 158–161.
6. E. SEIDEN, On a geometrical method of construction of partially balanced designs with two associate classes, *Ann. Math. Statist.* **32** (1969), 1177–1180.